

IS BASIC RESEARCH IN FIRMS USEFUL? EVIDENCE FROM PANEL DATA FOR SPAIN

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Abstract

We propose a model that associates basic research with product innovation through the absorptive capacity it generates, moderated also by the level of industry appropriability. To test these relations, we used a sample of 8,861 firms taken from a panel (PITEC). Our data suggest that basic research in firms increases their knowledge stock and flows, improving their capacity to identify, assimilate and exploit external knowledge and, therefore, their innovative product performance. By using lags we were able to find that this result can last for up to three years. In addition, we see that weak appropriability regimes can have a positive effect by increasing the influence of basic research on absorptive capacity. This evidence not only questions the disdain frequently shown by managers towards basic research, but also calls for open reflection on both the net effect of appropriability on innovative performance and the advisability of allocating public resources to stages and areas of the innovation process in which there are seldom ‘market failures’.

Key words:

Product innovation, basic research, absorptive capacity, appropriability.

¹ Contact author: xhvv@uvigo.es. We thank the Galician Department of Industry and Innovation and the Spanish Ministry of Science and Innovation for their economic support through projects 10SEC300036PR and ECO2008-03201/ECON respectively.

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1. INTRODUCTION

Practically throughout the 20th century science mostly took place in universities, while the responsibility for taking new or improved products and processes to the market was left almost exclusively to firms. However, this trend might have started to change in the early years of the 21st century (Pisano, 2010). The most recent data for Europe (2005) and for the USA (2007) suggest that the gap between private and university basic research is narrowing (Eurostat, 2011). The same source also confirms that in Japan basic research in firms is now even exceeding university research (42.7 € / inhabitant vs. 39.1 € respectively in 2007).

This change has undoubtedly arisen in part because universities have changed their budgetary focus from the ‘creation of science’ to the ‘business of science’, making greater efforts to utilize their intellectual property through licenses and spinoffs (Mowery et al., 2001; Gregorio and Shane, 2003; Sampat, 2006). However, the slowly-growing conviction among business managers about the profitability of investments in basic research has probably had a greater influence. This is something that has been recommended in academic circles for decades (Griliches, 1980; Mansfield, 1981; Rosenberg, 1990), and has given rise to empirical studies on its effects on productivity (Griliches, 1986), absorptive capacity (De Marchi and Rocchi, 2000), patents (Lim, 2004) and business profits (Henard and McFadyen, 2005).

In this context, the lack of empirical evidence to explain the link between basic research and product innovation is surprising. Take, for example, the most widely-quoted studies on new product development (Durisin et al., 2010) which make no reference at all to the influence of basic research, or the fullest meta-analysis on this subject (Henard and Szymanski, 2001) in which, out of the 24 variables for which sufficient correlations are found in the literature for them to be included in the study ($n > 10$ correlations), not one refers explicitly to investment in basic research. To cover this niche is, therefore, the basic aim of this paper.

This analysis also includes two additional contributions. Firstly, although the literature has emphasized the role of absorptive capacity in innovation processes, most studies focus on either its determinants (Jansen et al., 2005; Schmidt, 2010) or the moderating role played by certain variables (Oltra and Flor, 2003; Lim, 2004; Nieto and Quevedo, 2005; Tsai, 2009; Escribano et al., 2009). This moderating role traditionally granted to absorptive capacity suggests that it affects the sign and strength of relations between any input and the firm’s result in innovation. Lim (2004) suggests, for example, that general research activities are directly related to patent generation and that this relation becomes stronger with greater absorptive capacity. However, absorptive capacity might have an effect that conceptually

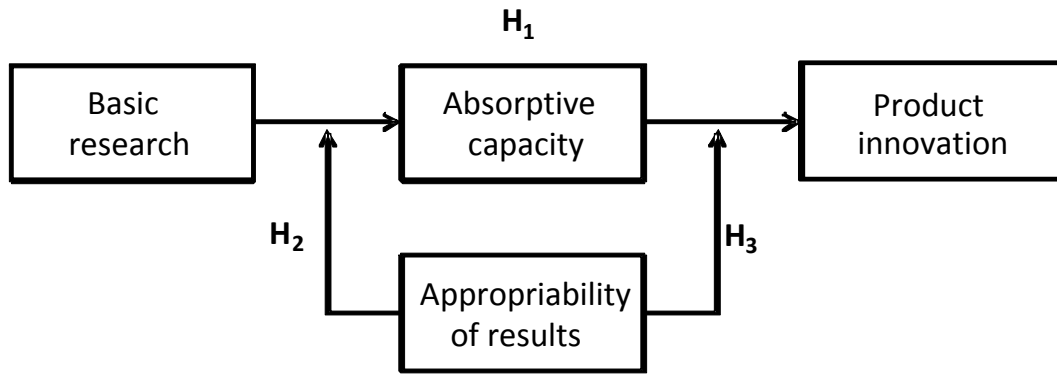
would be even more relevant if its influence were not just over the sign and strength of the relation but over its actual existence. This is what we call the mediating effect of absorptive capacity like the effect identified by Liao et al. (2010) between the acquisition of knowledge and global innovation capacity. Along the same lines, our study proposes a mediating role for absorptive capacity between basic research and product innovation.

Moreover, since the influence of appropriability on the background and results of absorptive capacity are still a matter of controversy (Todorova and Durisin, 2007), we include this variable in the model. In our paper we analyze the effect of appropriability, firstly, on the relation between basic research and absorptive capacity and, secondly, on the relation between absorptive capacity and product innovation. Our analysis suggests, and corroborates, that the absence of total appropriability of the results does not prevent the firms with most basic research from obtaining better results in product innovation.

Finally, from a strictly empirical viewpoint, the analysis has three strengths. Firstly, of special relevance is the use of lags in the relation between basic research and absorptive capacity. This allows us to test to what extent the basic research carried out in a given year affects the simultaneous or subsequent capacity for assimilating and exploiting the knowledge that becomes available. It can in fact be expected that, with today's fast technological change, the knowledge generated by basic research will soon be outdated. We therefore aim to identify how long the positive effect of basic research on future absorptive capacity can last. Secondly, in order to reduce any possible data bias and back up the widespread application of our results, we do not limit the sample to any specific sector. This is a relevant contribution because most studies on basic research focus on a specific sector, normally on knowledge-intensive sectors in which this type of research has traditionally been important, such as pharmaceuticals or biotechnology (Gambardella, 1992; Cockburn and Henderson, 1998; Lim, 2004). Finally, we build a multi-item index to measure absorptive capacity. By doing this we aim to reduce any measurement errors associated with the use of excessively simplifying proxies which, as with expenditure on R&D, do not reflect the multidimensional nature of the construct.

The paper is structured as follows: in section two we present the state of the art regarding the relations between basic research, absorptive capacity and appropriability in product innovation. Three hypotheses are drawn on which we build a model (Figure 1) to be tested in section four. In section three, we describe the sample of firms, the variables and the methodology used. In section 4 we cover the econometric analysis and discuss the results. The conclusions give our main findings and certain implications for product innovation theory, as well as recommendations for business management and for public policy.

Figure 1: Proposed model



2. THEORETICAL BACKGROUND AND HYPOTHESES

The study of product innovation determinants has occupied much of the research agenda over recent years, but there is still no consensus on the explanatory factors. The first studies were in the field of Industrial Economics and took their inspiration from Schumpeter, leaving a legacy that clarified matters significantly. So, although the evidence is ambiguous regarding the effect of firm size, there is a degree of consensus that certain characteristics of the sector such as technological opportunities (Levin et al., 1985; Geroski, 1990), market opportunities (Schmookler, 1966; Levin, 1981; Dougherty, 1990) or the conditions for appropriability of innovations (Mansfield, 1981, 1986; Levin et al., 1987) may influence product innovation differences between firms.

In parallel, studies carried out in the management field shared with Industrial Economics the emphasis on market characteristics (growth potential, rivalry, uncertainty) but they focused above all on stressing certain internal characteristics of organization (Vega-Jurado et al., 2008). They often adopted the Resource-based view of the firm (Wernerfelt, 1984), stressing the importance of generating distinctive competencies (Leonard-Barton, 1992; Tidd, 2000) for innovation strategy (Henard and Szymanski, 2001), technology (Love and Roper, 1999; Bhattacharya and Bloch, 2004), human resources (Hoffman et al., 1998; Song et al., 2003), organizational routines associated with management philosophies (Cooper, 1990; Rothwell, 1992; Souitaris, 2002; Webster, 2004) and the firm's absorptive capacity (Stock et al., 2001; Fosfuri and Tribó, 2008).

The role of basic research has been almost residual throughout the literature. Although there were some prior studies on the general effects of basic research (Mansfield, 1980; Griliches, 1986), it was probably Rosenberg (1990) who aroused interest amongst academics in the link between basic research and product innovation. Rosenberg suggested in this paper that basic research is essential for making strategic decisions on future product lines and the process technologies to be adopted. His argument was that the output of basic research is new knowledge which may be used for new product development. One year later, Pavitt (1991) completed this by pointing to the benefits of private basic

research – the development of certain skills, methods and networks of professional contacts allowing firms to better exploit external scientific conclusions. Implicitly, they both suggested, therefore, that basic research improves firms' absorptive capacity, that is, their capacity for identifying, assimilating and exploiting knowledge in their environment.

Perhaps because in 1990 the concept of absorptive capacity (Cohen and Levinthal, 1989) was not yet well-known, Rosenberg's argument was initially tested in some cases as a direct relation between basic research and patents (Gambardella, 1992). However, given the abstraction inherent in the results of basic research and the increasing importance of the absorptive capacity concept, it was not long before studies on basic research or product innovation started to consider the influence of absorptive capacity in one way or another. Generally speaking, we could classify these studies in two groups – those which study the influence of basic research on absorptive capacity, and those which focus on the relation between absorptive capacity and innovative performance.

The first group of studies suggests that firms that carry out their own R&D – either basic or applied – are in a better position to use any available external and internal information (Rocha, 1999; Kamien and Zang, 2000; Mancusi, 2004; Escribano et al., 2009). In particular, strictly with regard to basic research, several studies show a close relation with absorptive capacity (Lane and Lubatkin, 1998; Dyer and Singh, 1998; De Marchi and Rocchi, 2000), giving arguments that tie in exactly with what was stated by Rosenberg (1990) and Pavitt (1991) regarding the stock and flows of knowledge. So basic research was conceived as a learning process which, although not directly linked to the market, increases the depth and breadth of the stock of stored knowledge (Hernard and McFadyen, 2005) and consequently allows more to be learnt from any freely available internal and external information (Cassiman et al., 2002). In fact, the link between basic research and absorptive capacity is so close that even authors such as Cassiman and Veugelers (2006) use basic research as a measure of absorptive capacity.

Moreover, the literature on innovation performance emphasizes the capacity for using external knowledge sources as a key factor for innovation activities (Von Hippel, 1988; Tripsas, 1997; Cockburn and Henderson, 1998; Cohen et al., 2002; Chesbrough, 2003; Caloghirou et al., 2004; Rothaermel and Hess, 2007; Liao et al., 2010). In particular, since new product development is a knowledge-intensive activity that benefits from the acquisition and use of external scientific, technological or market information, greater absorptive capacity can be expected to improve product innovation (Stock et al., 2001). In fact, there is empirical evidence suggesting that firms with higher levels of absorptive capacity are more product innovative. For example, Stock et al. (2001) suggest that this relation has an inverted U shape, whereas Fosfuri and Tribó (2008) verify that firms with greater absorptive capacity systematically obtain greater percentages of sales of new, or substantially improved, products.

Clearly, all these studies have helped us to understand the close link between basic research, absorptive capacity and product innovation performance. It is surprising, however, that no studies have

aimed to check these relations as a whole, empirically exploring whether basic research has a positive effect on product innovation through absorptive capacity. The study which comes closest to this idea is by Lim (2004), who verifies empirically that absorptive capacity moderates the relation between research activities (both basic and applied) and invention capacity (number of patents). However, as the life of Benjamin Franklin illustrates clearly, capacity for invention is not the same as capacity for innovation. Moreover, granting a moderating role to absorptive capacity requires the implicit assumption that there is a direct link between basic research and product innovation, in which absorptive capacity only affects the strength of this link but does not explain why it exists. And assuming a direct link between basic research and product innovation is still surprising if we remember that basic research is defined as research that aims to further scientific knowledge without finding any specific marketable application. That is, obviously a firm may randomly generate a product innovation stemming from its basic research, but the proposal of a direct link between the two variables cannot arise randomly. Our proposal conceives absorptive capacity as a mediating variable between basic research and product innovation, because we consider this explains how and why such a link exists; that is, if there is no mediation by absorptive capacity, the link between these variables cannot be explained.

So, capacity for product innovation depends on the firm's capacity for recognizing the value of new, external information, assimilating it and applying it to commercial purposes (absorptive capacity), and this depends on the firm's level of prior knowledge – at least in part – stemming from the level of investment in basic research. Against this background, we pose the following hypothesis:

H₁: Firms' basic research increases product innovation by improving their absorptive capacity.

But, however strong the link between basic research, absorptive capacity and product innovation is statistically, considerations regarding the appropriability of the research findings may affect the strength of the link. In fact, the appropriability of innovations may have two basic effects on the initial model proposed: a first effect on the link between basic research and absorptive capacity, and a second effect on the link between the latter and the innovation result. In both cases, the literature points to clear areas of controversy that should be clarified (Todora and Durisin, 2007).

So, on the one hand, the more difficult it is for firms in a sector to appropriate new knowledge that they generate themselves, the greater the spillovers of knowledge becoming available for their competitors. The regime of industry appropriability will not only determine incentives for investing in the determinants of absorptive capacity (Cohen and Levinthal, 1990), but may also affect the actual strength of the relation between such determinants – in our case, basic research – and absorptive capacity. If the argument explaining this relation is based on the stock and flows of knowledge generated by basic research (Rosenberg, 1990; Pavitt, 1991), in weak appropriability regimes in which more, and better, external information is available, then firms are likely to have greater possibilities for improving their absorptive capacity. Firstly, because a higher level of available knowledge undoubtedly promotes the formation of human capital for the same level of investment in basic

research; and, secondly, because the additional interaction with other agents, which might arise from lower appropriability, will facilitate the additional identification of scientific and technological opportunities as well as the capacity for assimilating new knowledge that is scientifically far from what the firm had previously. We therefore expect that, the weaker the appropriability conditions, firms will be able to benefit more from a given level of basic research for generating absorptive capacity. When appropriability is strong, it will be the other way round. We therefore pose the following hypothesis:

H₂: Stronger appropriability regimes negatively moderate the relation between basic research and absorptive capacity.

On the other hand, the level of appropriability of innovations may also affect the relation between absorptive capacity and product innovation. In this case, however, there are two opposing forces which cast doubt on the net influence of appropriability. Zahra and George (2002) suggest that, in weak appropriability regimes, firms obtain less benefit from absorbed knowledge because other firms with complementary assets may gain at least part of the return from their innovations (Teece, 1986). This could be the interpretation of the empirical evidence presented by Escribano et al. (2009), who argue that the role of absorptive capacity is more relevant in environments characterized by strong protection of intellectual property rights. However, exactly the same argument could be used to explain why low appropriability of results may intensify the effect of absorptive capacity in product innovation. After all, just as a firm may lose returns from its absorptive capacity because of low appropriability of its results, the same firms could also be considered to be utilising their absorptive capacity even better to develop new products because they would be able to benefit from the knowledge generated by other firms. When the appropriability regime is strong, the opposite would happen. So, since we have diverging positions, we pose a hypothesis with two alternatives:

H_{3a}: The strongest appropriability regimes negatively moderate the relation between absorptive capacity and product innovation.

H_{3b}: The strongest appropriability regimes positively moderate the relation between absorptive capacity and product innovation.

3. DATA AND EMPIRICAL RESEARCH

3.1 Sample and measurement of variables

As our data base, we use the Technological Innovation Panel (PITEC), based on the Community Innovation Survey carried out in Spain under the leadership of the Spanish National Statistics Institute (INE). The PITEC has been monitoring technological activities in Spanish firms since 2004. It is therefore the result of stratified sampling in which four groups of firms are represented: firms with over 200 workers, firms with internal R&D expenditure, firms with less than 200 workers but with external R&D expenditure, and firms with less than 200 workers with no expenditure on innovation.

For this study we chose firms from this study that carried out innovation activity with a view to obtaining new products or processes during the period 2005-2007. So, although in 2007 13,291 questionnaires were sent out and 95.9% replies were received (INE surveys are obligatory by law), our sample comprises 8,861 firms, that is, 69.18% of the PITEC firms. Table 1 shows the main characteristics of the sample.

Table 1: Main characteristics of the sample

Characteristics	% of total
Business turnover:	
Up to 10 million	55.8
11-50 million	26.1
51 to 100 million	7.3
Over 100 million	10.8
Firm size:	
Up to 10 workers	13.3
11 to 50 workers	38
51 to 100 workers	15.1
101 to 200 workers	10.9
Over 200 workers	22.7
Sector:	
High and medium technology	38.8
Others	61.2

Drawn up by the authors based on PITEC for 2007

Table 2 describes the variables used in the study. In order to measure *product innovation*, we built a variable combining the importance of three effects that innovative activities may have on firms' products: a wider range of products and services, greater market penetration and higher-quality goods or services. Firms ranked these on a scale from 1 to 4, with 1 if the effect is slight and 4 if it is marked. This variable has been used in other studies, for example, Murovec and Prodan (2009).

For *absorptive capacity*, although the literature suggests different measurement methods, most studies use single-dimensional measures such as the firm's R&D activity (Cassiman and Veugelers, 2002; Oltra and Flor, 2003; Schoenmakers and Duysters, 2006; Nieto and Santamaría, 2007), the number of highly-trained workers (Caloghirou et al., 2004) or the importance of external sources of knowledge (Murovec and Prodan, 2009; Schmidt, 2010). However, these single-dimensional measures have been much criticized because they do not reflect the richness of the construct. In the process of transforming external knowledge flows into innovation results, the role played by absorptive capacity changes constantly and affects different capacities and routines at different times (Fosfuri and Tribó, 2008). Mangematin and Nesta (1999), Wong et al. (1999), Lane et al. (2001), Jansen et al., (2005), Nieto and Quevedo (2005), Vinding (2006), Tu et al. (2006), Liao et al. (2007), Escribano et al. (2009) and Camisón and Fores (2010) study absorptive capacity in greater depth and propose multi-dimensional operationalization of the construct.

In line with these authors, we propose multi-dimensional operationalization of the construct by bringing 5 variables together in an index generated by multiple correspondence analysis. The

synthesizing capacity of the resulting index is 72.8% of the variance and Cronbach's alpha is 0.907. The five variables covered by this index are the following:

- 1) Internal expenditure on R&D, because this expenditure not only generates new knowledge but also increases the firm's capacity for recognizing, assimilating and exploiting existing information, thus contributing to its absorptive capacity (Cohen and Levinthal, 1989). To measure this variable, we use firms' internal expenditure in 2007.
- 2) Systematic internal R&D activities, because firms which are constantly involved in such activities will have developed the necessary skills and experience to adapt external knowledge in their specific research fields so will have greater absorptive capacity (Becker and Peter, 2000; Oltra and Flor, 2003; Schmidt, 2010) than other firms which only occasionally carry out such activities. We measure this variable using a dummy taking value 1 if the firm carried out internal R&D in 2005, 2006 and 2007, and zero otherwise.
- 3) Workers' education and training, as this has a positive influence on firms' absorptive capacity (Rothwell and Dogson, 1991; Frenz et al., 2004). We measure this variable using the quotient between university-trained personnel in R&D and the total number of employees.
- 4) Training effort, because this also has a positive influence on employees' skill at assimilating external information (Frenz et al., 2004). To measure this variable, we use an ordinal variable taking 10 values depending on the percentage of sales allocated to training during the period 2005-2007.
- 5) The importance of external sources for innovation activities, because absorptive capacity can also be developed through links with external knowledge sources such as customers, suppliers, competitors, universities (Henderson and Cockburn, 1998) or research institutes (Mangematin and Nesta, 1999). This variable is measured using an index covering the importance of ten external sources (suppliers, customers, competitors, consultants, universities, public research bodies, technology centers, conferences, scientific journals and business associations) for innovation activities².

In addition, in order to quantify the *annual basic research* variable, we used the percentage of current internal expenditure on R&D that firms in our sample devote to this activity according to the PITEC panel for 2007, 2006, 2005 and 2004.

Finally, we included two control variables: *firm size*, because prior research has shown that innovation results might benefit from economies of scale and scope (Henderson and Cockburn, 1994; Escribano et al., 2009), measured here by the firm's turnover; and *sector of activity*, because this might determine the ease of producing innovations in terms of time and cost. In fact, it has been shown that relevant

² Firms rank the effects of these sources from 1 to 4: 1 if the source has not been useful for its innovation activities, and 4 if it has been very useful. This index is obtained by multiple correspondence analysis (Cronbach's alpha 0.89).

scientific and technological know-how advances at different speeds and with different degrees of difficulty (Klevorich et al., 1995).

Table 2: Description of variables

Variable	Designation	Description
INNOVPROD	Effects of the innovation on the firm's products	Sum of three effects caused by the innovative activities on the firm's products: (1) wider range of products and services; (2) greater market penetration, and (3) greater quality of goods or services.
AC	Absorptive capacity	Multiple correspondence analysis of the following variables: 1) internal expenditure on R&D; 2) systematic internal R&D activities; 3) university-trained personnel in R&D over total number of employees; 4) training effort made; 5) importance of external sources for innovation activities.
BR7	Basic research carried out in 2007	Percentage of current R&D expenditure in 2007 devoted to basic research.
BR6	Basic research carried out in 2006	Percentage of current R&D expenditure in 2006 devoted to basic research.
BR5	Basic research carried out in 2005	Percentage of current R&D expenditure in 2005 devoted to basic research.
BR4	Basic research carried out in 2004	Percentage of current R&D expenditure in 2004 devoted to basic research.
APROP	Industry appropriability	Total patents in the sector to which the firm belongs during the period 2005-2007 (grouped by the first two digits of the National Classification of Economic Activities).
SECTOR	Firm's sector	Takes value 1 if the firm belongs to a medium or high technology sector and value 0 otherwise.
SALES	Firm's turnover	Total firm's sales in 2007.

3.2 Analytical models

To test the existence of mediation as suggested in hypothesis H₁, the following regressions are necessary (Judd and Kenny, 1981; Baron and Kenny, 1986):

[1]

$$INNOVPROD = \beta_{10} + \beta_{11}BR7 + \beta_{12}BR6 + \beta_{13}BR5 + \beta_{14}BR4 + \beta_{15}SECTOR + \beta_{16}SALES + \varepsilon_1$$

[2]

$$AC = \alpha_{20} + \alpha_{21}BR7 + \alpha_{22}BR6 + \alpha_{23}BR5 + \alpha_{24}BR4 + \alpha_{25}SECTOR + \alpha_{26}SALES + \varepsilon_2$$

[3]

$$INNOVPROD = \beta_{30} + \beta_{31}BR7 + \beta_{32}BR6 + \beta_{33}BR5 + \beta_{34}BR4 + \beta_{35}AC + \beta_{36}SECTOR + \beta_{37}SALES + \varepsilon_3$$

where INNOVPROD is the firm's product innovation; BR7, BR6, BR5 and BR4 represent the basic research that firms carried out respectively in 2007, 2006, 2005 and 2004; AC is absorptive capacity and is the variable that mediates between basic research and production innovation; and SECTOR and SALES are the control variables in the model.

Following Judd and Kenny (1981) and Baron and Kenny (1986), in order for absorptive capacity to mediate between basic research and product innovation, four conditions are necessary. Since we

include investment in basic research over several years in the three equations (BR7, BR6, BR5 and BR4), we have to check whether these conditions apply in the basic research carried out every year, just some years or not at all.:

1. The total effect of basic research on product innovation should be significant; that is, the following parameters should be significant in equation [1]: β_{11} for basic research in 2007, and β_{12} , β_{13} , and β_{14} for 2006, 2005 and 2004 respectively.
2. The effect of basic research on absorptive capacity (the mediating variable) should be significant; that is, the following parameters should be significant in equation [2]: α_{21} , α_{22} , α_{23} and α_{24} .
3. The effect of the absorptive capacity mediating variable on the product innovation variable should be significant, that is, the β_{35} coefficient in equation [3] should not be zero.
4. The residual effect, that is, the effect after discounting the indirect mediating effect, of basic research on product innovation should be lower in absolute value than the total effect, that is, $|\beta_{31}| < |\beta_{11}|$ in 2007; $|\beta_{32}| < |\beta_{12}|$ in 2006; $|\beta_{33}| < |\beta_{13}|$ in 2005; and $|\beta_{34}| < |\beta_{14}|$ in 2004.

To test the role played by appropriability in both the relation between basic research and absorptive capacity (H_2) and the relation between absorptive capacity and production innovation (H_{3a} and H_{3b}), we follow the process described by Muller et al. (2005), so we consider the following equations:

[4]

$$AC = \alpha_{40} + \alpha_{41}BR7 + \alpha_{42}BR6 + \alpha_{43}BR5 + \alpha_{44}BR4 + \alpha_{45}APROP + \alpha_{46}APROP * BR7 + \alpha_{47}APROP * BR6 + \alpha_{48}APROP * BR5 + \alpha_{49}APROP * BR4 + \alpha_{410}SECTOR + \alpha_{411}SALES + \varepsilon_4$$

[5]

$$INNOVPROD = \beta_{50} + \beta_{51}BR7 + \beta_{52}BR6 + \beta_{53}BR5 + \beta_{54}BR4 + \beta_{55}APROP + \beta_{56}APROP * BR7 + \beta_{57}APROP * BR6 + \beta_{58}APROP * BR5 + \beta_{59}APROP * BR4 + \beta_{510}AC + \beta_{511}APROP * AC + \beta_{512}SECTOR + \beta_{513}SALES + \varepsilon_5$$

where, in addition to the previous variables, the APROP variable appears referring to industry appropriability and the interaction between this variable and the basic research carried out in the different years and absorptive capacity. Note that, following the recommendations given by Muller et al. (2005), we center all variables at their mean. This facilitates interpretation of the parameters as given in Table 3.

Table 3: Interpretation of the parameters involved in the hypotheses

Equation [1]	
β_{11} β_{12} β_{13} β_{14}	Total effect of basic research each year (2007, 2006, 2005 and 2004, respectively) on product innovation, that is, the sum of possible direct and indirect effects.
Equation [2]	
α_{21} α_{22} α_{23} α_{24}	Effect of basic research each year (2007, 2006, 2005 and 2004, respectively) on absorptive capacity.
Equation [3]	
β_{31} β_{32} β_{33} β_{34}	Residual direct effect of basic research each year (2007, 2006, 2005 and 2004, respectively) on product innovation controlled by absorptive capacity, that is, after discounting the indirect effect exerted via absorptive capacity.
β_{35}	Effect of absorptive capacity on product innovation.
Equation [4]	
α_{41} α_{42} α_{43} α_{44}	Effect of basic research each year (2007, 2006, 2005 and 2004, respectively) on absorptive capacity for an average level of industry appropriability.
α_{45}	Effect of industry appropriability on absorptive capacity for an average level of basic research in the different years.
α_{46} α_{47} α_{48} α_{49}	Change in the effect of basic research each year (2007, 2006, 2005 and 2004, respectively) on absorptive capacity as a result of an increase in industry appropriability.
Equation [5]	
β_{51} β_{52} β_{53} β_{54}	Direct residual effect of basic research each year (2007, 2006, 2005 and 2004, respectively) on product innovation for an average level of industry appropriability.
β_{55}	Effect of industry appropriability on product innovation for an average level of investment in research in the different years and for an average level of absorptive capacity.
β_{56} β_{57} β_{58} β_{59}	Change in the direct residual effect of basic research each year (2007, 2006, 2005 and 2004, respectively) on product innovation as a consequence of increased industry appropriability.
β_{510}	Effect of absorptive capacity on the firm's product innovation at an average level of investment in basic research in the different years and at an average level of industry appropriability.
β_{511}	Change in the mediating effect of absorptive capacity on product innovation as a consequence of increased industry appropriability.

So, for there to be moderation by appropriability, at least one of the two following conditions must be present. These need to be verified, as in the previous case, in the basic research carried out each year.

Condition 1:

The following parameters in Equation [4] must be significant: α_{46} , α_{47} , α_{48} and α_{49} as well as parameter β_{510} in Equation [5].

Condition 2:

The following parameters in Equation [4] must be significant [4]: α_{41} , α_{42} , α_{43} and α_{44} as well as parameter β_{511} in Equation [5].

If the first of the conditions is met, hypothesis H₂ will be corroborated. If both conditions are met, then either H_{3a} or H_{3b} will be corroborated, depending on the sign of the coefficients.

4. RESULTS

This section shows the results of the Equations [1] to [5] proposed by Muller et al. (2005). In order to control for the effects we are interested in, the control variables for turnover and sector of activity have been included in all the models.

Table 4: Direct and indirect effects of basic research and mediation by absorptive capacity

Variable	Absorptive capacity (Mediating)	Product innovation	
	Equation [2]	Equation [1]	Equation [3]
Basic research 2007	0.014***	0.013**	-0.004
Basic research 2006	0.009***	0.011**	0.000
Basic research 2005	0.003***	0.006**	0.002
Basic research 2004	0.002***	0.002	0.000
Absorptive capacity			1.212***
Sector	0.551***	1.048***	0.380***
Figure	1.155e-10***	1.418e-10**	1.827e12
Constant	-0.015	0.005	0.022
R ²	0.111	0.038	0.196

Table 4, which shows the coefficients resulting from estimating Equations [1], [2] and [3], also indicates that the four conditions necessary for there to be mediation are met:

- (1) There is a direct, significant, positive effect of basic research most of the years (2007, 2006 and 2005) on product innovation (Equation [1]). The fact that basic research in 2004 did not have effects on product innovation can be explained because the larger stock of knowledge provided to the firm might have become outdated because of the high speed of technological change .
- (2) There is a positive, significant effect of basic research every year (2007, 2006, 2005 and 2004) on absorptive capacity (Equation [2]).
- (3) There is an important positive, significant effect of absorptive capacity (mediating variable) on product innovation (Equation [3]).
- (4) The direct residual effect of basic research in 2007, 2006 and 2005 on product innovation (Equation [3]) is lower in absolute value than the direct effect (Equation [1]). In fact, variables BR7, BR6 and BR5 are no longer significant so there is total mediation by absorptive capacity between them and product innovation. The residual effect of basic research in 2004 does not need to be checked because, as already stated, this variable showed no direct effect on product innovation, that is, the first condition was not met.

The above results confirm that the basic research carried out in 2007, 2006 and 2005 had a positive effect on product innovation via the absorptive capacity generated in the firm. In other words, our data empirically support hypothesis H₁.

In order to test the moderating role of the appropriability regimes in the above relation, hypotheses H₂ and H_{3a}/H_{3b}, we use the coefficients for Equations [4] and [5], which are given in Table 5. However, since basic research in 2004 had no effects on product innovation for the period, we leave out the role of appropriability for basic research that year.

Table 5: Moderation by the appropriability of results

Variable	Absorptive capacity (Mediating)	Product innovation
	Equation [4]	Equation [5a]
Basic research 2007	0.020***	0.001
Basic research 2006	0.011***	0.003
Basic research 2005	0.003***	0.001
Industry appropriability	0.000***	-6.193e-5
Basic research 2007 * industry appropriability	-1.765e-5***	-1.506e-5
Basic research 2006 * industry appropriability	-1.118e-5**	-2.706e-5
Basic research 2005 * industry appropriability	1.567e-6	4.958e-7
Absorptive capacity		1.226***
Absorptive capacity * industry appropriability		0.000
Medium-high technology sector	0.427***	0.410***
Turnover	1.145e-10***	9.214e-12
Constant	0.002	0.011
R ²	0.097	0.184

The results corroborate the moderating role of appropriability regimes only in the relation between basic research and absorptive capacity (H₂), because condition 1, as explained above (Muller et al., 2005), is only met for the investment in basic research in 2007 and 2006. That is, the coefficients for Equation [4] α_{46} , and α_{47} and parameter β_{510} in Equation [5] are significant. Therefore, the knowledge spillovers generated in 2007 and 2006 help increase the stock and flow of knowledge generated by firms by investing in basic research by means of a higher level of absorptive capacity in them. The fact that appropriability regimes do not moderate this relation for basic research in 2005 (α_{48} is not significant) may be because the greater knowledge in 2005 generated by spillovers in weak appropriability regimes is no longer so relevant. Because of the speed and dissemination of technological change, such knowledge, which is available to all, may have become outdated so is of limited usefulness in 2007 for firms' human capital, for their relation with other agents or for identifying new business opportunities.

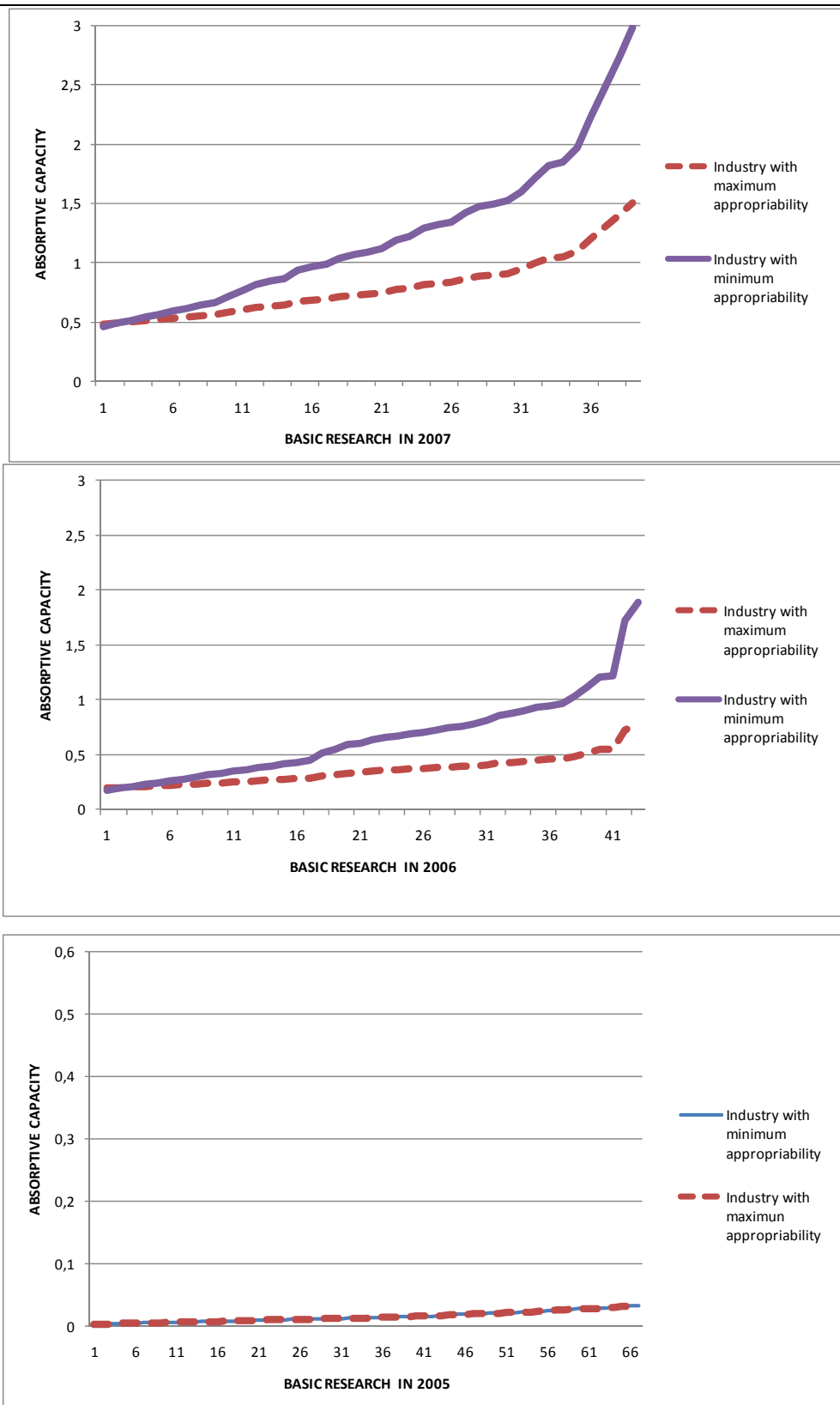
With regard to hypotheses H_{3a} and H_{3b}, our data seem to rule out any moderating effect by appropriability on the relation between absorptive capacity and product innovation. Our explanation for this is the opposing effect of the two hypotheses. Some of the literature suggests that, in weak appropriability regimes, firms might obtain less benefit from absorbed knowledge because other firms capture part of the return from their innovations. But other literature suggests that the low appropriability of results might intensify the effect of absorptive capacity on product innovation by making it easier for firms to use knowledge generated by others. Perhaps what is happening is that the first argument is applicable to firms that follow offensive innovation strategies while the second is correct for firms that follow imitative strategies (Freeman, 1974). Since we have no information on

such strategies, studying the phenomenon in an aggregate way might lead the statistical effect to be zero.

In Figure 2 we show the effects of industry appropriability (moderating variable) on the relation between basic research each year and absorptive capacity. In the first graph, on basic research for 2007, three effects can be observed. Firstly, basic investment exerts a positive influence on absorptive capacity in both strong and weak appropriability regimes (both curves have a positive slope). Secondly, this positive influence is greater in firms operating in weak appropriability regimes (the curve representing the sectors with maximum appropriability has a steeper slope). And, thirdly, the beneficial effect of weak appropriability regimes is greater with higher levels of investment in basic research (with greater basic research, the curves become increasingly separate).

Analysis of the second graph, the one for basic research in 2005, also shows these three characteristics. Moreover, if we compare this graph with the previous one, we see that basic research for 2006 has less influence on absorptive capacity in 2007 (both curves slope less than the same curves in the 2007 graph). Finally, the third graph reveals that the positive influence of basic research in 2005 on absorptive capacity in 2007 is not affected by the level of industry appropriability (both curves have the same slope). In addition, the influence of basic research in 2005 on absorptive capacity is lower than that of basic research in subsequent years (the curve again has less of a slope than in the previous cases).

Figure 2: Effects of the industry appropriability regime on the relation between basic research and absorptive capacity



5. CONCLUSIONS

Our results confirm that basic research improves product innovation through the effect it has on absorptive capacity. This happens because, even though basic research by definition does not seek a specific marketable application, it does increase the stock and flow of knowledge needed to extract, assimilate and exploit under the best conditions the latest scientific and technological developments which may turn into new or improved products.

Moreover, it should be stressed that the effect of basic research on absorptive capacity and, therefore, on product innovation has a time limit. That is, however much a firm invests in basic research during any given year, the effect on its absorptive capacity will have a limited duration unless it perseveres. If its effort is not systematic, fast-moving scientific and technological change will lead the knowledge generated by the investment in basic research to quickly become outdated. We have seen in this study that the positive effect of investing in basic research might last up to three years.

We also verified that the effect of basic research on absorptive capacity is greater in firms that belong to sectors with weaker appropriability regimes. This might be due not only to the fact that a greater level of spillovers in the environment would stimulate the formation of human capital for the same level of investment in basic research, but there might also be an additional interaction in that environment between different agents that can improve the identification of new scientific and technological opportunities.

We have not, however, been able to test empirically whether the level of appropriability affects in any way the relation between absorptive capacity and product innovation. A possible future line of research could study whether this result indicates that comparing these arguments would require more information on the firm's innovation strategy. In weak appropriability environments, it may be that the effect of absorptive capacity on product innovation in firms with offensive innovation strategies is limited because other firms take advantage of their effort. Similarly, perhaps other firms with imitative innovation strategies might be able to take advantage of exactly this, thus making the effect of their absorptive capacity on their product innovation even more powerful.

Our results have certain implications for business management and for public policy. On the one hand, managers should ignore passing fads of little intellectual depth to question conventional thinking regarding basic research and product innovation. Short-sightedness focusing exclusively on market-based R&D, neglecting the generation of human capital amongst their technical staff can only lead to a gradual loss of capacity to support substantial product innovations. In particular, regarding basic research as a source of advantages associated with the stock and flow of knowledge, our results should qualify some of the most important drawbacks faced by managers when investing in basic research, mainly because the desired long-term effects of basic research in fact do not last so long. We have in fact found that the investment in basic research carried out by firms in 2005, 2006 and 2007 affects absorptive capacity and product innovation during that same period. Secondly, our analysis, carried out on all types of firms from all sectors and of all sizes, generates results that are very predictable

regarding product innovation capacity. By predictability we are not referring to the specific improvements they will make but to the fact that we have found a robust relation between basic research and improvements of various types in firms' products. Finally, the low appropriability of basic research (which has nothing to do with the above-mentioned industry appropriability regimes) is the third main challenge for business management. Our results indicate two things. Firstly, the arguments used here to justify basic research in firms have little to do with its appropriability, that is, its specific, immediate, exclusive, return. Once its importance has been justified by its effects on absorptive capacity (a tacit dynamic capacity), arguments on the difficulty of converting basic research into specific market innovations are of little use. Moreover, if what managers are concerned about is that the results of basic research might generate spillovers to other firms, then they might wish to resort to industrial secrecy, as is being done by a growing number of firms to protect the results of their applied research (Pissano and Teece, 2007; Ottoz and Cugno, 2008). But, anyway, firms' researchers do not need academic publications to gain promotion.

Regarding public policy, our results have two possible implications. Firstly, since public funding is limited, financial aid should not focus on projects that are so close to the market that public stimulation is practically irrelevant for implementing them. Such projects generate interest *per se* and hence it is more difficult to justify public intervention (Arrow, 1962), which under these circumstances often ends only reducing costs at the cost of distorting competition. Things are different, however, with projects where there is research, particularly basic research (Nelson, 1959). In this case there are clear market failures which the designers of public policies can try to resolve by supporting knowledge-intensive projects.

We are not of course suggesting that public policies should aim to only promote basic research in firms but, just as administrations appreciate universities coming closer to the market, the designers of public policies should do everything possible to ensure that firms take care to generate knowledge stock and flows amongst their technical staff. Otherwise, not only will regions find that their technologists become increasingly good in increasingly obsolescent areas, techniques and procedures, but they will also lose opportunities to generate intra and inter-sector spillovers by providing the only human capital that can solve problems at the forefront of knowledge (Zellner, 2003).

Finally, a second implication of our study for public policy relates to the appropriability of innovation results. Traditionally, it has been suggested that strong systems to protect results encourage investment in research and development, so regulators should strengthen industrial and intellectual property rights. One intriguing result of our study, however, indicates that strong systems of industrial and intellectual property also lead to a smaller effect of basic research on firms' absorptive capacity. This might indicate that needs differ depending on the country's technological capacity. So, international pressure to design strong appropriability systems might benefit only the most technologically-advanced countries. On the other hand, the countries with the greatest difficulties from the point of view of technology might require their industrial property rights to be strengthened at a different rate; perhaps

they should first strengthen their firms' absorptive capacity and, only when investment in R&D is systematic and guaranteed, should legislators focus their attention on strengthening the appropriability of innovations.

6. BIBLIOGRAPHY

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